

4/ Use the following terms to complete the following sentences. : (5 Marks)

A) instruction set	B) USB flash drive	C) BIOS ROM	D) Clock speed
E) SATA/PATA	F) Electromagnetic interference (EMI)	G) CMOS	H) Electrostatic discharge (ESD)

1. 3. Hard drives and CD/DVD-ROM drives connect to the power supply with a _____.
2. The _____ is the rate at which a CPU executes instructions.
3. The BIOS system configuration is stored on nonvolatile memory called _____.
4. The movement of a static charge from you to something you touch is called _____.
5. A solid-state device commonly used to store data is called a _____.

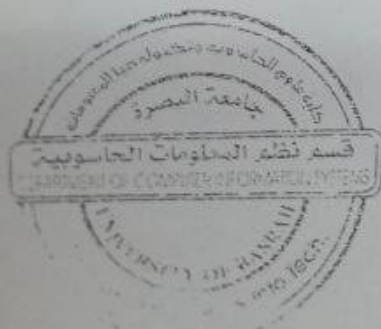
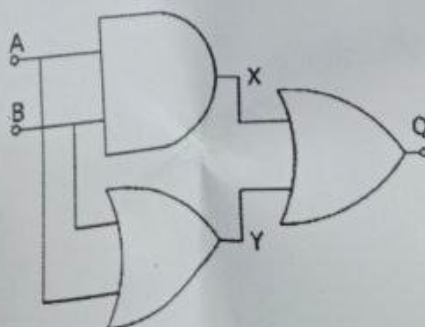
Q5/ Write the number of sentences and the symbol (✓) for the correct sentence and the symbol (×) for the incorrect one of the following sentences: (5 Marks)

1.	Serial and Parallel ports are the latest type of printer interface.
2.	The removable memory sticks commonly used in today's cameras and PDAs use programmable read-only memory or PROM
3.	Software applications that run on computers with CPU chips made by Intel will also run on Apple computers that have CPU chips made by Motorola.
4.	Devices that are attached, either via a cable or wireless connection, to a given computer are called peripherals .
5.	ISA is a type of expansion slot that is almost always used for high-speed, 3D graphics video cards.

Q6/ Answer the following

(5 Marks)

1- Find the output of the following logic circuit :



A	B	(A.B) + (A+B)
0	0	
1	0	
0	1	
1	1	

2- Convert the system numbers:

$$(101010101)_2 = (\quad)_{10}$$

END OF TEST

Examiner

Asst. Prof. Asaad Al-Hijaj

Head of the Department

Prof. Dr. Haider M. Mashhadi

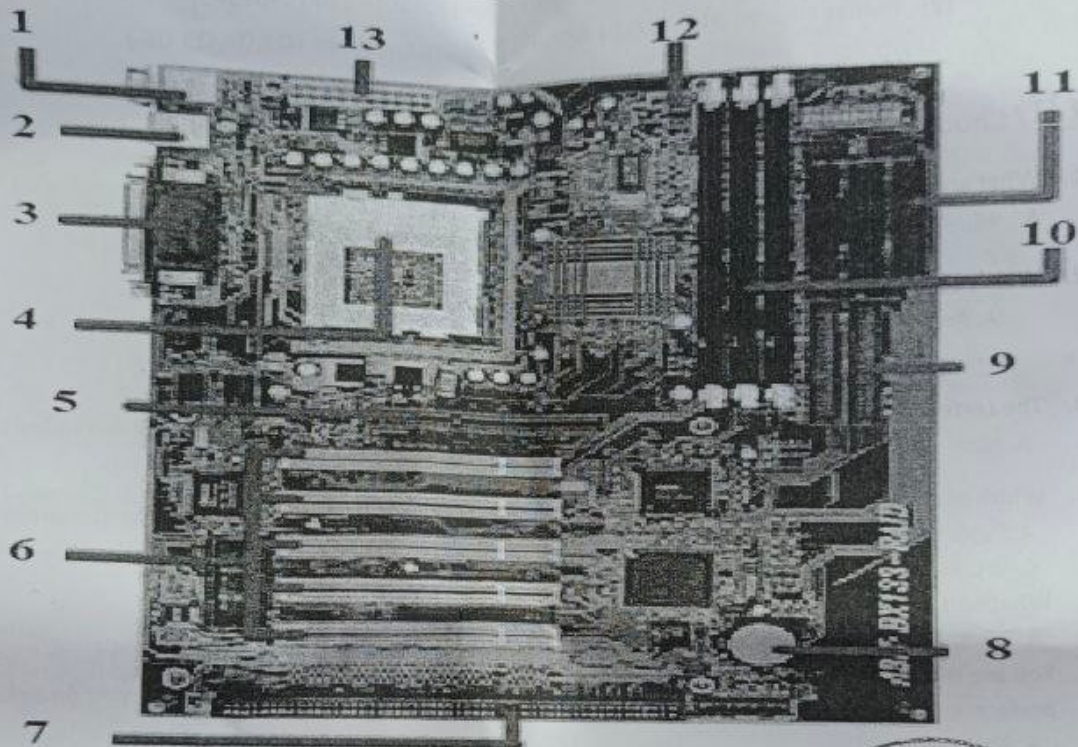


The Dean



Q2/ Match 10 of the terms in table below with the following figure : (10 Marks)

A) CPU Fan Plugin	B) Power Supply Plugin	C) Keyboard & Mouse	D) USB	E) Parallel Port
F) CPU Socket	G) AGP Socket	H) PCI Slot	I) ISA Slot	J) CMOS Battery
K) Memory Slot	L) Floppy Connector	M) IDE Connector (Hard, CD-ROM)		



Q3/ Answer four of the following questions.:

(5 Marks)



- 1) There are some important *Windows system management tools*. Count and explain one?
- 2) There are several characteristics and technologies that can affect a *CPU's performance*. Count and explain one?
- 3) Why digital *multi-meters* are one of important tool for computer technician?
- 4) Explain briefly, What is the **POST** and what's their functions?



Final Exam of Computer Fundamentals
University of Basrah
College of Computer Science and Information Technology
Department of Computer Information Systems
2024-2025 / First Attempt



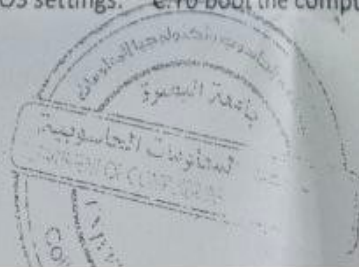
Exam Duration: 2 Hours

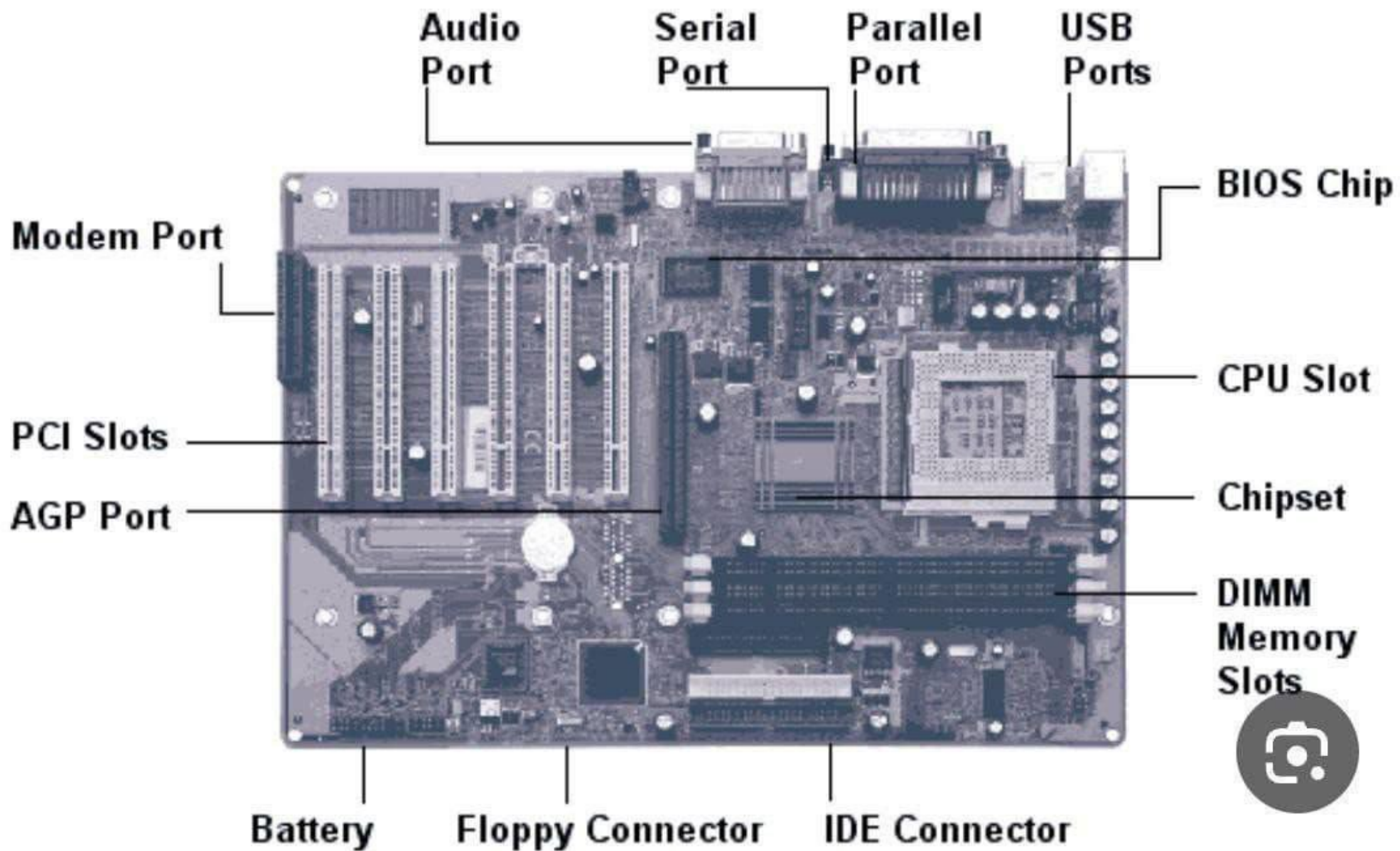
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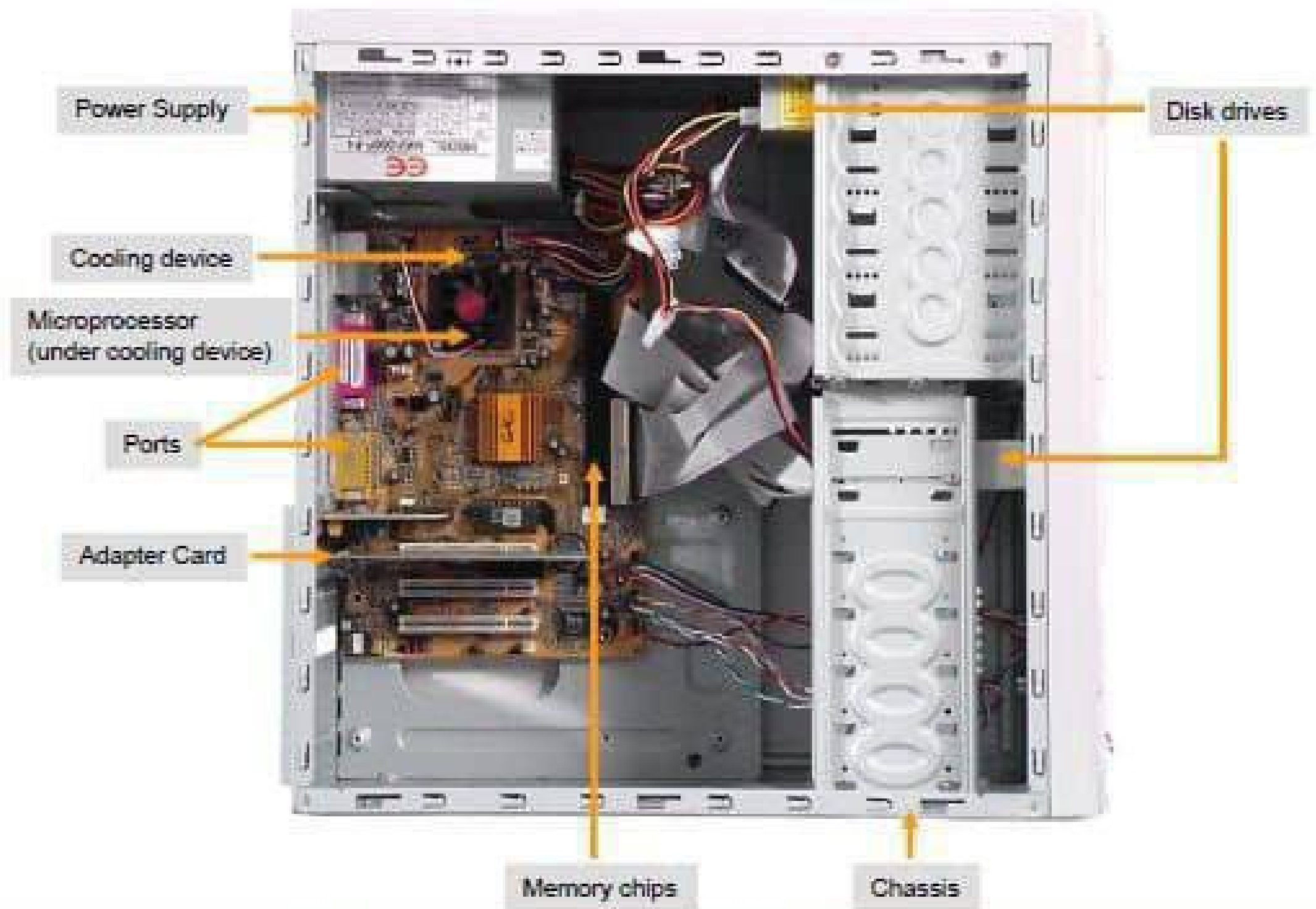
- N.B. (1) The maximum marks for this paper is $35 = [10+10] + 3 \text{ of } [5+5+5+5]$
(2) (Must answer Q1 and Q2) Answers 3 of the questions (Q3,Q4,Q5,Q6)

Q1/ Choose the correct answer of the following sentences: (10 Marks)

- What are the 4 types of DC voltage usually produced by a power supply?
A. +4, -4, +16, -16 B. +3, -3, +9, -9 C. +5, -5, +12, -12 D. +4, -5, +6, -3
- Which of the following could cause a computer to automatically reboot?
A. An overheating CPU. B. A five second power failure.
C. Insufficient Memory. D. A new set of software installed.
- The correct connection method for an internal laptop hard disk drive would be which of the following?
A. SCSI B. PATA C. IEEE 1394 D. Firewire
- Which of the following tools is a place to get a quick look of what is being shared on your computer?
A. Performance Logs and Alerts B. Disk Management
C. Local Users and Groups D. Shared Folders
- What sort of connector does a typical network card have?
A. DB-9 B. Mini-DIN C. RJ-11 D. RJ-45
- You are the technician charged with performing a hardware upgrade on a PC. What will you do before performing the procedure?
A. Back up all the data. B. Determine the amount of RAM available.
C. Scan the hard disk for errors. D. Download the Latest BIOS and drivers for the existing components.
- Which type of expansion slot is almost always used for high-speed, 3D graphics video cards?
A. USB B. AGP C. PCI D. ISA
- When installing a new CPU, what is applied directly to the CPU?
A. Cooling liquid. B. Thermal compound. C. A heat sink. D. A fan
- When you add new hardware devices to Windows, it will detect the new hardware. What do Windows use to detect the new hardware?
A. Plug and Play B. Add New Hardware Wizard C. Device Manager D. Control Panel
- What is the CMOS memory used for?
A. Keeping the time B. To store BIOS settings. C. To boot the computer D. None of these options is correct









Final Exam (First attempt) of Academic year 2025-2026 Second Semester (Copy 1)

Notice – (1) The questions are on three pages. (2) Five marks each questions.
(3) Answers must be written in pencil only. (4) Answer only 7 Seven questions.

Q1 Choose the Correct answer of the following sentences:

(5 Marks)

1.	Which of the following components on the motherboard usually house the IDE connectors? A. PCI bus. B. North Bridge. C. ISA bus. D. South Bridge.	2.	How can you tell that a PC has successfully completed its Power on Self Test (POST)? A. Two beeps at startup B. A single beep at startup C. A light on the motherboard comes on D. The computer turns on
3.	What happens when the CMOS battery fails? A. All the CMOS configuration information is lost B. The computer won't boot C. The computer is destroyed D. The motherboard is dead	4.	Which of the following is a characteristic of the FAT file system? A. Supports compression, encryption, disk quotas, and file ownership. B. Allows for file-level security to protect system resources. C. Compatible with DOS and Windows 9x dual-boot configurations. D. None of the above
5.	What is the purpose and function of the SCANDISK command? A. Allows you to examine the hard drives of the computer B. Allows you to delete files and subdirectories and is a more powerful extension of the DEL command C. Allows you to duplicate floppy disks D. Allows you to delete a file from the folder	6.	When you start your computer, it performs an important test. What is this test known as? A. CPU Processing B. A CMOS test C. POST D. Hard drive test
7.	To dispose of laptop batteries, what would you suggest the correct procedure would be? A. Incineration. B. First try to recharge the batteries for reuse. C. Consult the local guidelines for disposing of hazardous material. D. Dispose in a trash can.	8.	Which of the following is the first thing you should do when replacing a PCI card? A. Shut down the system and unplug the power B. Open the case C. Open the package and sit it beside the computer D. Purchase the device
9.	What is the CMOS memory used for? A. Keeping the time B. To store BIOS settings C. To boot the computer D. None of these options is correct	10.	Which of the following memory module form factors are commonly used for laptops? A. Dual Inline Memory Modules (DIMMs) B. Small Outline DIMMs (SODIMMs) C. Single Inline Memory Modules (SIMMs) D. Rambus Inline Memory Modules (RIMMs)



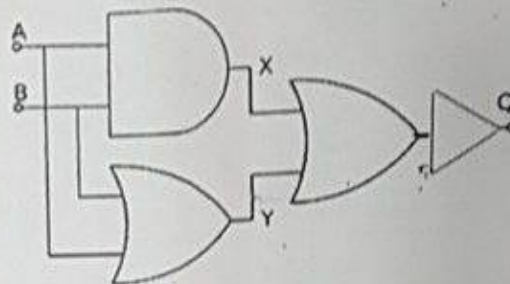
Q6/ Answer Two of the following questions:

(5 Marks)

- 1) There are various types of hard disk drives. List out.
- 2) There are various PC connection methods (Ports). List out.
- 3) There are various types of RAM Memory. List out.

Q7/ Find the output of the following logic circuit :

(5 Marks)



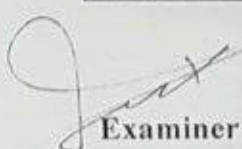
A	B	$AB = X$	$(A+B) = Y$	$X + Y$	Q
0	0				
0	1				
1	0				
1	1				

Q8/ Convert the system numbers for Two of the following:

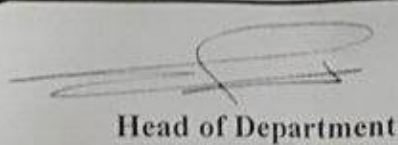
(5 Marks)

- 1- $(2A5C)_{16} = (\quad)_{10}$
- 2- $(0001110001)_2 = (\quad)_{10}$
- 3- $(325)_{10} = (\quad)_2$

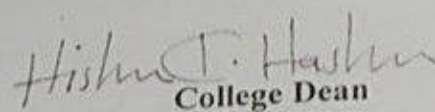
With best wishes for success


Examiner

Prof. Asaad A. Al-Hijaj


Head of Department

Asst. Prof. Biada AbdulQader


College Dean

Prof. Dr. Husham Kadhum Hashim

Q2/ Answer the following questions: (Short Answer)

(5 Marks)

1. What is the CPU speed units , give examples ?
2. What are the four basic units of electricity? Provide the unit name and symbol.
3. What type of expansion slot is compatible with the video adapter card?
4. What is the key or combination of keys used to enter the BIOS setup program?
5. Why would you change the first boot device to the optical drive?

Q3/ Answer the following questions: (Full Explain)

(5 Marks)

1. Why is an NTFS partition more secure than FAT32?
2. Why is a digital multimeter an important piece of equipment for a technician? Explain your answer.
3. List and describe levels of permissions on shared folders.

Q4 / Write the symbol (✓) for correct sentence and the symbol (✗) for incorrect one of the following sentences:

(5 Marks)

1. (Hard drive, RAM, ROM) are used to store the data and programs for repeated use? Information can be added and deleted at will, and it does not lose its data when power is removed.
2. Devices that are attached, either via a cable or wireless connection, to a given computer are called peripherals.
3. ISA is a type of expansion slot that is almost always used for high-speed, 3D graphics video cards.
4. Partitioning is the process that is used when you are assigning part or all of the drive for use by the computer.
5. Green and Purple are the color codes used for the keyboard Port on the old PC that uses a mini-DIN 6 Connector.

Q5/ Define 4 of the following terms: (Briefly)

(5 Marks)

- | | | |
|------------------|------------------|-----------------|
| 1) EEPROM | 2) BNC connector | 3) Cache memory |
| 4) Plug and Play | 5) CMOS | 6) Hot swapping |





Final Exam of Information Technology Essentials
University of Basrah
College of Computer Science and Information Technology
Department of Computer Information Technology
2022-2023/ Second Semester



Exam Duration: 3 Hours

Date: 11/06/2023

- N.B.
- (1) The maximum mark for this paper is 35
 - (2) The marks for each question are shown on the **right side** of the question.
 - (3) You are expected to provide an example where appropriate.
 - (4) Answers to questions should be grouped and written together, i.e., all responses to sub-questions of individual questions like Q1, Q2, Q3, etc., should be answered one after the other.
 - (5) Please read each question carefully before you start to answer it.

Q1: Write the **number** of the sentence and the **letter** of the correct answer for (9) sentences: 9
[1x9]

- 1) The _____ is a type of external device.
a) CPU b) memory c) HDD **d) scanner**
- 2) _____ is a graphical interface that provides access to a number of utilities that you can use to configure the Windows operating system or the computer's hardware.
a) My Documents b) Windows Explorer **c) Control Panel** d) Start Menu
- 3) A _____ is a special connector used for diagnosing transmission problems that redirect electrical signals back to the transmitting system.
a) multimeter b) share c) toolkit **d) loopback plug**
- 4) _____ enables you to enter text-based commands or run command-line tools.
a) Windows **b) Command Prompt** c) Unix d) Linux
- 5) _____ is a personal computer connection that transfers data eight bits at a time over eight wires and is typically used to connect a printer to a system unit.
a) parallel connection b) USB c) SCSI d) serial connection
- 6) _____ is a personal computer connection that provides a high-speed interface for peripheral devices that are designed to use the IEEE 1394 standard.
a) parallel connection **b) FireWire connection** c) SCSI d) serial connection
- 7) There are various types of hard disk drives such as _____.
a) USB 2.0 **b) Parallel ATA** c) Berg connector **d) FDD**



- 8) _____ use to automatically configure the card's hardware resources.
a) AMR (Audio Modem Riser) **b) I/O address** c) MIDI devices d) PnP
- 9) Optical drives include such as _____ drives.
a) FDD b) HDD **c) DVD** d) CMOD
- 10) There are several types of disk partitions such as _____.
a) Media file systems b) NTFS c) FAT32 **d) Primary**

Q2: Define only Five of the following terms:

- 1) Firmware 2) System bus 3) Hot swapping 4) Device driver 5) Plug and Play
6) BIOS

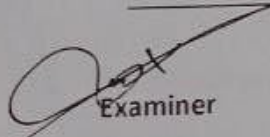
Q3: Answer Two of the following (Briefly):

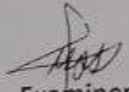
- 1) There are various types of hard disk drives. List out.
2) There are various PC connection methods (Ports). List out.
3) There are various types of RAM Memory. List out.

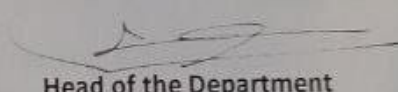
Q4: Answer Two of the following:

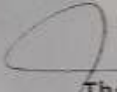
- 1) What is the Operating System? Explain its functions.
2) There are several characteristics and technologies that can affect a CPU's performance. Count and explain.
3) There are some important Windows system management tools. Count and explain.

END OF TEST


Examiner
Asst. Prof. Asaad Al-Hajjaj


Examiner
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Heidar J. Fadhil


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Asst. Prof. Dr. S

